

Allan Paiz

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SKILLS

- **Languages:** C++, Python, Java, Bash
- **Libraries:** Pandas, NumPy, JUnit, Plotly, Matplotlib, Scikit-learn
- **Robotics & Simulation:** ROS 2, Gazebo, TurtleBot4, mobile robots, multi-agent path finding
- **Tools:** LaTeX, Neovim, Git, GitHub, Linux, Unreal Engine 5, Rider, Excel, PowerPoint, Photoshop, Premiere Pro

EDUCATION

Bachelor of Science in Computer Science - University of South Carolina Aug 2023 - Dec 2026
| Columbia, SC | 3.7 GPA | Minor in Mathematics

| **Academic Honors:** Fall 2023, Spring 2024, Spring 2025, Fall 2025

Relevant Coursework: Software Engineering, Computer Architecture, Advanced Programming Techniques, Data Structures & Algorithms, Discrete Mathematics, Artificial Intelligence, Math Foundations of ML/Data Science, Optimization, Linear Algebra, Operating Systems, Transition to Advanced Mathematics

EXPERIENCE

National Science Foundation Computer – Research Experience for Undergraduates May 2026 - Jul 2026
Computing Research Association / University of South Carolina - Columbia, SC

- I wrote a technical report proposing a ROS 2 perception architecture for TurtleBot4, integrating OAK-D stereo vision, 2D LiDAR, object detection, tracking, sensor fusion, and safety aware control; demonstrated the pipeline with YOLOv8 and LiDAR trajectory tests.
- I explored the mathematical foundations of robotic perception including projective geometry, stereo-depth estimation, LiDAR processing, and Kalman filtering to formulate uncertainty aware obstacle detection and tracking.
- I developed Gazebo simulation software for control of multi-agent path finding using ROS 2 and TurtleBot4 robot models.

Undergraduate Research Assistant Jan 2026 - May 2026
University of South Carolina, Department of Computer Science and Engineering - Columbia, SC

- I assisted with research under Dr. Vignesh Narayanan in the Dynamical Systems and AI Research Group.
- I ran hardware tests with TurtleBot4 robots using ROS 2 and collected experimental data for mobile ground robot control research.
- I worked on control of mobile ground robots, including obstacle-avoidant navigation, constrained optimization, and event-triggered optimal control.

Advanced Data Collector May 2024 - Present
Pro Football Focus - Remote

- I collect NFL and NCAA football player data focusing on speed, precision, and consistency. I have processed 23 NCAA and 5 NFL games identifying and recording over 180,000 data points.
- In the 2024 season I achieved a 99.5% player identification accuracy, 96.7% player position accuracy, and a 98.9% accuracy across all other categories, validated through PFF's double-entry verification process.

US Marine May 2018 - Oct 2022
United States Marine Corps - Camp Pendleton, CA

- I used computer based geographic information systems (GIS) to create and analyze maps and spatial data for mission planning and reconnaissance reporting.
- I served as a Sapper Leaders Course Instructor, mentoring and teaching Marine leaders through lecture and hands on training in mission planning, problem solving, and small unit leadership.
- I worked effectively in a variety of teams and challenging environments, including overseas deployments and advanced technical training.

Technical Designer Jun 2015 - Dec 2016
Production Resource Group - New Windsor, NY

- I collaborated with clients and project managers to engineer solutions for entertainment scenery and machinery projects, meeting strict safety, functional, budget, and timeline constraints.

- I used AutoCAD for 2D and 3D modeling, drafted technical drawings for manufacturing, and created assembly instructions with a bills of materials following company standards.

PROJECTS

Wash Your Rice

August 2025 - May 2026

Unreal Engine Game - Group Project

- **Links:** [Project Link] [Capstone Page]
- **Focus:** Unreal Engine 5.6.1, C++, Rider, game development, collaboration
- **Description:** Worked in a team of five to develop a horror themed puzzle game built in Unreal Engine 5. You play as a student trapped in an overnight nightmare. To survive until morning, you solve computer based puzzles, manage your sanity, react to enemy attacks, and uncover codes that provide an escape.

Construction Permit Delay Risk Prediction

Jan 2026 - May 2026

Math Foundations of ML/Data Science Course Project

- **Focus:** Python, Scikit-learn, SVM, neural networks, TF-IDF, data preprocessing, classification
- **Description:** Used open source Atlanta building permit data to identify high risk construction projects with a higher risk of experiencing delays for permits to be issued.
- Built and compared a kernel SVM baseline, neural network classifiers, and a neural network classifier with description text features to predict normal versus high risk permit delay categories.

Event-Triggered Optimal Control with Safety Guarantees for Obstacle-Avoidant Navigation of Nonholonomic Robots

Jan 2026 - May 2026

Optimization Course Project

- **Focus:** Optimization, nonholonomic wheeled mobile robots, constrained path planning, event-triggered control, control barrier functions, neural networks
- **Description:** Presented a unified framework for obstacle-avoidant trajectory planning and resource-efficient control of nonholonomic wheeled mobile robots.
- Formulated path planning as a constrained nonlinear optimization problem and used an adaptive neural network-based critic to approximate the Hamilton-Jacobi-Isaacs equation for event-triggered optimal control.

Defensive Stopping Power

Oct 2023 - Jan 2024

2024 NFL Big Data Bowl - Kaggle Competition

- **Links:** [Project Link] [Finalists Link]
- **Focus:** Python, machine learning, data analysis, visualizations, feature engineering, self-directed
- **Description:** I developed a feedforward neural network in Python to detect contact between ball-carriers and defenders using NFL player tracking data. I created a novel metric, Yards Allowed After Contact, to evaluate defensive performance for player scouting and game analysis.

TuneUp by JETA

Jan 2025 - Apr 2025

Software Engineering Course - Team Project

- **Links:** [GitHub Link] [Demo Video]
- **Focus:** Java, version control, collaboration, documentation
- **Description:** Worked as a team to develop a music application written in Java. I worked in an agile, iterative development process implementing design patterns, object oriented programming, and unit testing. I created documentation, system design diagrams, and used GitHub for version control.

AWARDS

NFL Big Data Bowl - Runner Up

2024

My submission was selected as one of the top 10 finalists, earning a runner up prize of \$5,000. [Kaggle Announcement] [NFL Announcement]

Marine Corps Good Conduct Medal

2021

Outstanding performance and conduct during three years of continuous active enlisted service in the U.S. Marine Corps.

Certificate of Commendation

2021

Outstanding performance of duty while serving as a seminar leader, Lance Corporal Leadership and Ethics Seminar Class 3-21.